
Weini Huang

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Scientific career and Education

- 01/2018- **Lecturer in Mathematical Biology**, School of Mathematical Sciences, Queen Mary University of London.
Principal Investigator, Theoretical Biology Lab, College of Life Science, Sun Yat-Sen University, Guangzhou, China.
Research topic: tumour and resistance evolution, predator-prey & host-parasite coevolution, stability of ecological communities.
- 03/2016- **Postdoc**, Evolution and Cancer Laboratory, Centre for Tumour Biology, Barts Cancer
12/2017 Institute, Queen Mary University of London.
Research topic: theoretical modelling of tumour dynamics, evolution of drug resistance in ovarian cancer.
- 11/2012- **Postdoc**, Department of Evolutionary Theory, Max Planck Institute for Evolutionary
01/2016 Biology, Plön, Germany.
Research topic: stochastic models on growing populations, impact of trade-offs on population dynamics under coevolution.
- 10/2008- **Dr. rer. nat.**, Research Group for Evolutionary Theory, Max Planck Institute for
08/2012 Evolutionary Biology, Plön, Germany.
Thesis: Evolutionary game dynamics driven by mutations under frequency dependent selection.
- 09/2005- **M.S.**, Ecology, Beijing Normal University, Beijing.
07/2008
- 09/2002- **B.A.**, English, Beijing Normal University, Beijing (Minor).
03/2005
- 09/2001- **B.S.**, Information System and Technology, Beijing Normal University, Beijing (Major).
07/2005

Publications

- **W. Huang***, A. Traulsen, B. Werner, T. Hiltunen, L. Becks, 2017. Dynamical trade-offs arise from antagonistic coevolution and decrease intraspecific diversity. *Nature Communications*, 8:2059.
- A. Baker, **W. Huang**, et al., 2017. Robust RNA-based in situ mutation detection delineates colorectal cancer subclonal evolution. *Nature Communications*, 8:1998.
- A. Amado, L. Fernández, **W. Huang**, F. F. Ferreira, P. R. A. Campos*, 2016. Competing metabolic strategies in a multilevel selection model. *Royal Society Open Science*, 3:160544.
- **W. Huang***, P. R. A. Campos*, V. M. de Oliveira, F. F. Ferreira*, 2016. A resource-based game theoretical approach for the paradox of the plankton. *PeerJ*, 4:e2329.
- **W. Huang**, C. Hauert, A. Traulsen*, 2015. Stochastic game dynamics under demographic fluctuations. *Proceedings of the National Academy of Sciences*, 112:9064-9069.
- A. Amado, **W. Huang**, P. R. A. Campos, F. F. Ferreira*, 2015. Learning process in public goods games. *Physica A: Statistical Mechanics and its Applications*, 430:21-31.
- **W. Huang**, B. Werner, A. Traulsen*, 2012. The impact of random frequency-dependent mutations on the average fitness. *BMC Evolutionary Biology*, 12:160.
- **W. Huang**, B. Haubold, C. Hauert, A. Traulsen*, 2012. Emergence of stable polymorphisms driven by evolutionary games between mutants. *Nature Communications*, 3:919.
- X. Luan*, G. Sun, Y. Qu, **W. Huang**, D. Li, S. Liu, B. Wu, 2012. Prioritizing biodiversity in conservation planning based on C-Plan: a case study from northeast China. *Acta Ecologica Sinica*, 32(3):715~722 (In Chinese with English abstract available).
- **W. Huang**, A. Traulsen*, 2010. Fixation probabilities of random mutants under frequency dependent selection. *Journal of Theoretical Biology*, 263(2):262~268.
- X. Luan*, **W. Huang**, D. Li, M. Liu, 2009. Identification of hotspots and gaps for biodiversity conservation in the Northeast of China based on a systematic conservation planning methodology. *Acta Ecologica Sinica*, 29(1):144~150 (In Chinese with English abstract available).
- L. Zhang*, F. Liu, **W. Huang**, J. Song, 2008. Occurrence of Nematode parasites in raptors in Beijing, China. *Journal of Raptor Research*, 42(3):204~209.

* corresponding authors

Reviewer Activities

Journal of Theoretical Biology, Physical Review Letters, Physical Review E, PNAS, Proc. Royal Soc. B, Proc. Royal Soc. Interface, PLoS Computational Biology, Nature Communications, American Naturalist, BMC Evolutionary Biology, Annals Of Oncology, Gene, Evolution Letters, Evolutionary Applications
FWO, Research Foundation - Flanders in Belgium

Awards and scholarships

- 2016 The 6th Integrated Mathematical Oncology Workshop "Resistance", Research team award, Moffit Cancer Center, Tampa, US
- 2016 1000 talented youth program, Ministry of Education, China
- 2014 Best presentation award, ECMTB 2014, European Conference on Mathematical and Theoretical Biology, Gothenburg, Sweden.
- 2012-2015 Postdoc fellowship, Max Planck Institute, Germany
- 2008-2012 PhD scholarship, China Scholarship Council
- 2008-2009 International fellowship, American Association of University Women Educational Foundation

Organized workshops

- 07/2017 Foresting cancer evolution symposium, Mathematical Models in Ecology and Evolution, London, Germany.

10/2010 First workshop Gen-AG 'evolutionary genetics': from genes to ecosystems, German Society of Genetics, Kappelsberg, Germany.

Presentations

Invited

- 05/2018, 15th Annual Internal Symposium at the Max Planck Institute for Evolution Biology - Aquavit 2018, Plön, Germany (Guest scientist in the best presentation award committee; Talk: Evolutionary dynamics with tradeoffs).
- 04/2018, Xiamen University, School of Mathematics, Xiamen, China (Talk: Spatial patterns in cancer evolution).
- 12/2017, East China Normal University, School of Environmental Science, Shanghai, China (Talk: From species coevolution to cancer evolution).
- 10/2017, DynaTrait International Conference, Hannover, Germany (Talk: Dynamical trade-offs under coevolution).
- 03/2015, Graduate school of Gulbenkian Institute of Science, Lisbon, Portugal (Mini-course: Evolutionary Game Theory as one part of the Evolution course, 1 lecture, 3h/lecture)
- 10/2014, Center for Models of Life, Niels Bohr Institute, Copenhagen, Denmark (Talk: Stochastic game dynamics in populations with changing size).
- 02/2014, Summer school of Department of Physics, Federal University of Pernambuco, Recife, Brazil (Mini-course: Evolutionary Game Theory, 3 lectures, 1.5h/lecture)
- 05/2011, International symposium for biodiversity and theoretical ecology, Guangzhou, China (invited talk: Polymorphisms emerges from evolutionary mutant games).

Others

- 12/2017, The International Society for Evolution, Ecology and Cancer, Tempe, US (Talk: Exploring the evolution of cancer and therapy resistance by theory and experiments).
- 07/2017, Mathematical Models in Ecology and Evolution, London, UK (Talk: Revealing mutant fitness and timing by spatial mixing of sub-clones in tumour).
- 07/2016, Evolutionary Biology and Ecology of Cancer Summer School, Cambridge, UK (poster).
- 05/2015, Modelling Biological Evolution: Linking Mathematical Theories with Empirical Realities, Leicester, United Kingdom (Talk: Modelling frequency dependent interactions between types under demographic fluctuations).
- 06/2014, European conference on mathematical and theoretical biology, Gothenburg, Sweden (Two talks: 1. Evolutionary game dynamics under demographic fluctuations; 2. A resource-based model of multi-species competitions under mutations).
- 08/2013, Models in Population Dynamics and Ecology, Osnabrück, Germany (Talk: Stable polymorphisms driven by evolutionary games between mutants).
- 03/2013, 77th annual meeting of the DPG (German physics society) and DPG spring meeting, Regensburg, Germany (Talk: Evolutionary game dynamics under random mutations).
- 07/2011, Summer school on dynamical models in life sciences organized by European society for mathematical and theoretical biology, Evora, Portugal (poster).
- 09/2009, Evolution of cooperation: models and theories, Laxenburg, Austria (poster).

Field work experiences

- 08/2006-10/2006 Population survey of elephants in the tropical forests of Xishuanbanna, Yunnan Province, China
- 07/2006-08/2006 Bird survey in the north-west mountain area in Sichuan Province, China

Other experiences

- 10/2007 Conservation International, Investigation of illegal trade of Chinese Pangolin (*Manis pentadactyla*) in Jiangxi province, China
- 04/2005-06/2006 Roots & Shoots (The Jane Goodall China), Volunteer & group leader of animal welfare and room enrichment team, Beijing Zoo

09/2005- Beijing Raptor Rescue Center, Volunteer & keeper
02/2006

2004 WWF/ Friend of Nature, Volunteer & environment investigator